

VAMSI UDAYAKUMAR JONNAKUTI

Brooklyn, NY | +1 914-494-5812 | vj2280@nyu.edu | linkedin.com/in/vamsiuk | github.com/VamsiUdaykumar

EXPERIENCE

Target Corporation

Aug 2022 – Aug 2024

Senior Software Engineer

- Architected the transition of a low-code automation platform to a unified no-code one, designing the **Postgres** database and **Java/Spring** microservices, saving **65,000 man-hours/year** for over **500** internal users.
- Spearheaded the creative automation capabilities using **OCR & ML** to extract design components along with relative positions to render package designs on Adobe Illustrator using **Java & ActiveX**, with **80%** workload reduction.
- Built a color extraction pipeline that processed artwork files using ML and then automated updates to **SPARK via Kafka-based services**, eliminating **100+ hours/year** of manual work at a rate that is **70% faster**.
- Led the migration from **Nomad to TAP** by revamping CI/CD pipelines, developing a web driver framework, and a credential management API, improving system stability and reducing downtime by **40%**.
- Prototyped an Intelligent Document Processing (IDPaaS) system by integrating **OCR** with **text extraction** to enable automation of semi-structured document processing and reducing manual review effort.
- Trained and mentored 6 team members in **Java, Spring Boot, Vela, and Docker**; guided 2 automation projects that improved expertise in the tech stack and agile SDLC.

Target Corporation

Dec 2019 – Aug 2024

Software Engineer

- Developed five robust, scalable bots using **Selenium and Java**, automating key business processes that saved over **\$500,000** annually while reducing manual effort by **2,000 hours** across two distinct business units.
- Revamped the manual intervention capability on the monitoring dashboard using **ReactJS and Redux**, helping users update data and change the processing state in real-time, reducing failure scenarios by **75%**.
- Built a supervised text classification pipeline for ServiceNow tickets using NLP preprocessing and **Logistic Regression with TF-IDF** to automate ticket routing, reducing manual triage effort and resolution time by **30%**.

PROJECTS

TwinMind Pro - Live Meeting Copilot | *React, Web Audio API, Groq, Whisper, Llama-3*

- Engineered a real-time AI meeting assistant using **React** and **Groq API**, implementing a custom **Web Audio API** and heuristic noise filters to process continuous audio streams and deliver sub-second actionable insights.

Agent Runs API & Observability | *Python, FastAPI, OpenTelemetry, Grafana, SQLite, Server-Sent Events (SSE)*

- Engineered an API in **Python** to orchestrate multi-step AI agent executions, integrating an event-sourced **SQLite** backend, SSE for live streaming, and **OpenTelemetry/Grafana** for distributed tracing.

AGNews LoRA Fine-Tuning | *Python, HuggingFace Transformers, PEFT, LoRA, PyTorch*

- Fine-tuned a **BERT-based** model on AGNews using **LoRA** and **HuggingFace PEFT**, achieving **92.25% accuracy** with reduced trainable parameters and faster convergence.

Medical Chatbot MLOps | *Docker, Kubernetes, GitHub Actions, ArgoCD, FastAPI, MLFlow*

- Developed an MLOps pipeline with **Argo Workflows/CD** for auto-retraining and CI/CD of a transformer-based medical chatbot deployed on a **Kubernetes cluster** provisioned via **Terraform** and **Ansible**.

EDUCATION

New York University

Sep 2024 – May 2026

Master of Science, Computer Engineering

TECHNICAL SKILLS

- **Languages:** Python, Java, C/C++, C#, JavaScript, TypeScript, SQL, HTML/CSS, Bash
- **Frameworks & Libraries:** Selenium, Angular, .NET, Groovy, Maven, Gradle, ReactJS, Spring Boot, Flask, FastAPI, GraphQL, REST APIs, PyTorch, Scikit-learn, HuggingFace, LoRA, PEFT, MLflow, Ray, Pandas, NumPy, OpenCV, JUnit, Mockito
- **Tools & Technologies:** Docker, Kubernetes, Terraform, Jenkins, Nomad, Git, Linux, ArgoCD, Ansible, NoSQL, PostgreSQL, MongoDB, MySQL, Kafka, Prometheus, Grafana